

Leakage-Aware, Interpretable Feature Fusion for Automatic Short Answer Grading across Heterogeneous Datasets

Ashraf Kahlout

Affiliation — to be added

Abstract

Automatic short answer grading (ASAG) systems are increasingly accurate but opaque, and their reported accuracy is often inflated by an under-examined evaluation artifact: when answers to the same *question* appear in both training and test folds, models exploit question-specific shortcuts rather than genuine answer understanding. We present (i) a systematic **question-leakage audit** across six heterogeneous ASAG datasets, (ii) an interpretable feature-fusion grader—a NaN-native gradient-boosted head over lexical, semantic, and rubric-coverage feature branches—and (iii) an explainability study whose attributions are validated against pre-existing human gold feedback. Replacing the still-common stratified k -fold protocol with grouped leave-questions-out cross-validation collapses apparent performance by up to 38 macro-F1 points (Powergrading $0.830 \rightarrow 0.450$), and a control model that sees *only* the question identity nearly matches the full model under the leaky protocol (0.873 vs. 0.830). Under the honest protocol, our tuned head beats trivial and question-shortcut baselines on four of six datasets, with significance established by a cluster bootstrap over questions under Holm correction—a procedure that reverses two apparent wins that an item-level bootstrap would have accepted. Branch ablations show hand-crafted linguistic features remain the accuracy workhorse, while the rubric branch is *faithful but accuracy-neutral*: its concept-coverage signals rise monotonically with human verdicts on the SAF corpus (AUC up to 0.62), providing teacher-style explanations at no accuracy cost. We further report calibration (temperature scaling halves SemEval ECE, $0.122 \rightarrow 0.059$), perturbation robustness, and leave-one-dataset-out transfer. A DeBERTa cross-encoder arm, fused as an out-of-fold feature that preserves exact TreeSHAP attributions, is evaluated under identical unseen-question folds ([**TBD: hybrid headline sentence after Colab run**]). All code, splits, and reports are released.

Keywords: automatic short answer grading; educational NLP; evaluation methodology; data leakage; explainable AI; SHAP; calibration; gradient boosting; transformers

1 Introduction

Short answer grading asks a model to score a free-text student response against a question and, usually, a reference answer. The task sits between paraphrase detection, textual entailment, and ordinal regression [1, 2], and a deployed grader must be both accurate *and* auditable by the teacher who signs the grade. Two problems motivate this work.

Evaluation honesty. Several widely used ASAG corpora (Mohler [3], Powergrading [4], MIND-CA [5]) ship without official train/test splits, so practitioners fall back on stratified k -fold cross-validation over answers. But if one question’s answers are distributed across folds, the model can memorize question-specific regularities—the answer distribution, typical phrasing, even answer

length—and these shortcuts [6] vanish on genuinely new questions. The community has long known that unseen-question generalization is harder (SemEval-2013 defined unseen-answer/question/domain test sets a decade ago [2]; Condor et al. showed the drop directly for SBERT graders [7]), yet the leaky protocol remains common practice, and *how much* it inflates published-style numbers on these standard corpora has not been quantified. We quantify it, with a diagnosis (between-question variance share), a control (a model that sees only the question identity), and a remedy (grouped leave-questions-out CV).

Interpretability. High-capacity encoders grade well but cannot tell a teacher *why* a mark was given, and post-hoc explanations of black-box graders are rarely validated against human judgment [8, 9]. A fusion grader whose inputs are human-readable (overlap, negation, concept coverage) admits exact per-feature attributions via TreeSHAP [10, 11]; the question is whether those attributions actually track what human graders reward. The SAF corpus [12] uniquely ships per-answer human verification feedback, which we use as a *pre-existing* gold standard for explanation validation —no new annotation required.

Research questions.

- **RQ1** (leakage): How much does question-level leakage inflate ASAG results under stratified k -fold CV, and what remains under grouped leave-questions-out evaluation?
- **RQ2** (branches): Which interpretable feature branches (semantic, linguistic, rubric-coverage) carry the accuracy, and what do negation-scope cues add?
- **RQ3** (explanations): Are the head’s explanations faithful to the model, and do the interpretable signals align with human gold feedback?
- **RQ4** (deployment): How calibrated and perturbation-robust is the grader, and does anything transfer across datasets?
- **RQ5** (hybrid): Does an out-of-fold transformer signal, fused as one more feature, improve unseen-question accuracy while preserving exact attributions? [**TBD: answered after the Colab run**]

Contributions. (1) A question-leakage audit across six heterogeneous ASAG datasets, pairing a seen-question upper bound with an honest grouped protocol and a question-shortcut control baseline (Section 5, Table 2); (2) an interpretable, NaN-native feature-fusion grader evaluated across classification, ordinal, and regression targets with per-dataset Optuna tuning that never touches test data (Section 4); (3) branch ablations isolating each feature family’s contribution under the honest protocol (Table 5); (4) an explainability study combining exact TreeSHAP, a faithfulness decomposition, rubric concept attribution, and validation against SAF’s human gold feedback (Section 7.5); (5) a cluster-bootstrap significance analysis with Holm correction that reverses two apparent wins (Table 4); (6) calibration, robustness, and leave-one-dataset-out transfer analyses (Sections 7.6–7.8); and (7) a hybrid arm fusing out-of-fold DeBERTa signals into the same interpretable head under identical folds (Section 7.9, [**TBD: pending**]). Code, splits, and all report artifacts are released for reproduction.

2 Related Work

2.1 Automatic short answer grading

ASAG has evolved from hand-engineered similarity features [3, 13, 1] through the SemEval-2013 shared task [2] to neural scorers [14] and transformer fine-tuning [15, 16, 17]. Surveys note that the strongest reported systems often combine hand-crafted features with transformer semantics [18], and a scoping review finds the marginal contribution of embeddings over conventional features inconclusive [19]—both observations motivate keeping an interpretable feature core. Similarity-based (reference-comparison) scoring has been defended as more classroom-suitable than instance-based scoring because it transfers to new questions with few labels [20, 21]; our rubric and semantic branches follow that design. ASAG2024 recently unified seven corpora into a combined benchmark [22]; we share the multi-dataset spirit but keep each dataset’s native target and metric, adding protocol-level auditing instead of scale unification.

2.2 Evaluation protocols and question leakage

SemEval-2013 institutionalized unseen-answer (UA), unseen-question (UQ), and unseen-domain (UD) test sets [2], and SAF ships UA/UQ splits [12]. Condor et al. compared random vs. question-level holdouts for SBERT-based ASAG and observed sharp drops [7]; recent datasets quantify the same cliff [23]. In essay scoring, Li and Ng re-examined cross-prompt evaluation and found simple feature-based models match neural state of the art once prompts are properly held out [24]. Our audit differs in three ways: it targets the *no-official-split* ASAG corpora where stratified k -fold remains the default, it reports the inflation explicitly as a seen-vs-grouped gap with a question-identity-only control model, and it pairs the audit with variance diagnostics (η^2) that predict which datasets are most leakage-prone. To our knowledge no prior work publishes such an audit for Mohler, Powergrading, and MIND-CA. Shortcut learning is the general framing [6].

2.3 Explainable ASAG

Post-hoc attribution methods have been applied to transformer graders [8], and ExASAG generates natural-language explanations validated against recruited experts [9]. AERA distills LLM rationales into a smaller scoring-plus-explaining model [25]. Our approach is complementary and cheaper: the model is interpretable *by construction* (31 human-readable features), the attributions are exact rather than approximate [11], and validation uses SAF’s *pre-existing* human verification feedback rather than newly collected judgments—turning a shipped dataset field into a free gold standard for explanation quality.

2.4 Hybrid feature–transformer graders

GradeAid concatenates TF-IDF features with BERT cross-encoder similarity for regression heads across public ASAG corpora [26]; hybrid feature-plus-embedding ensembles are likewise common in essay scoring. Our hybrid differs in the evaluation and explanation contract: the transformer signal enters as an *out-of-fold* prediction (never fit on its own test fold), the fusion head remains NaN-native gradient boosting with exact TreeSHAP, and neural-only / feature-only / hybrid arms are compared under identical grouped unseen-question folds. **[TBD: tighten this paragraph once hybrid numbers exist]**

Dataset	Domain	Task / target	Questions	Answers	Split regime
SemEval-2013 T7 [2]	science tutoring	5-way clf / label	252	16,003	official UA/UQ/UD
SAF [12]	comm. networks	regression / score	31	2,981	official UA/UQ
ASAP-SAS [37] [†]	science, biology	ordinal / QWK	4 prompts	8,722	official train/dev/test
Mohler [3] [‡]	CS data structures	regression / Pearson	21	616*	grouped CV
Powergrading [4]	US civics	binary clf / macro-F1	20	4,941*	grouped CV
MIND-CA [5]	psychology (children)	ordinal / QWK	11	10,543*	grouped CV

Table 1: Evaluation datasets. *after within-question deduplication. [†]via the AERA mirror [25]: EssaySets 1, 2, 5, 6 (4 of 10 prompts) but with gold test-split scores. [‡]via the ASAG2024 subset [22]: 21 of the original 80 questions. Because Mohler and ASAP-SAS are subsets, we report internal numbers only and make no head-to-head claims against published results (Section 7.10).

2.5 LLM-era grading

Zero- and few-shot LLM graders are attractive but currently unreliable as sole graders: they trail fine-tuned or classical supervised systems on agreement metrics [27], and GPT-4-with-prompting has been reported to lose to traditional models in recent head-to-heads [28]. Rationale-distillation improves over raw ChatGPT scoring [25]. Our contribution is orthogonal: an audited evaluation protocol and a validated, CPU-only, transparent head that institutions can run locally; the protocol itself applies unchanged to LLM graders.

2.6 Robustness, calibration, and deferral

Adversarial studies show ASAG systems can be fooled by universal triggers and even adjective/adverb insertion [29, 30, 31], motivating perturbation testing. Confidence estimation for routing uncertain answers to humans was proposed for Japanese SAS [32] and extended to human-in-the-loop cost analyses [33]. We follow temperature scaling [34] for calibration and report risk-coverage curves; our significance methodology follows the cluster (block) bootstrap [35] with Holm-Bonferroni control [36].

3 Datasets and Task Formulation

Every loader emits one unified schema (`question_id`, `question`, `reference_answer`, `student_answer`, `score`, `label`, `dataset`, `domain`, `split`), enforced by tests. The six datasets (Table 1) deliberately span three task types—5-way classification, ordinal, bounded regression—three split regimes, and five subject domains. Two datasets (ASAP-SAS, MIND-CA) provide no reference answer, so 24 of 31 reference-dependent features are intentionally NaN there and ignored natively by the tree head (no imputation). For corpora without official splits, exact duplicate answers within a question are collapsed to the median-score row before splitting (Powergrading 13,960 $\rightarrow$ 4,941; MIND-CA 11,311 $\rightarrow$ 10,543; Mohler 1,260 $\rightarrow$ 616), removing a trivial memorization vector. Licensing and acquisition details for each corpus are documented in the released repository; raw data are not redistributed.

4 Method

4.1 Two-view preprocessing

Each row is normalized into two parallel views written to separate columns. The *encoder view* applies NFKC and whitespace normalization only, preserving the signal transformers expect. The *feature view* applies spaCy [38] lemmatization, lowercasing, and punctuation/stopword removal *with negators preserved*. A third, negation-scope view additionally prefixes tokens inside a negator’s scope (window of 4 kept tokens, closed early at clause boundaries), so that “does not forward” yields `neg_forward`; contrasting features built from the plain vs. scope-marked views provides a ready-made negation ablation (Section 7.4). An empirical subword token-length study over both a WordPiece and a SentencePiece tokenizer justified `max_len = 128` for the encoder regime ($\leq 1.09\%$ truncation on five datasets), with a per-dataset override of 256 for SAF, whose multi-point reference solutions are long.

4.2 Interpretable feature branches (31 features)

Features are grouped into three branches, each a pure function of the two views:

- **Branch A — semantic (3):** SBERT [39] (all-MiniLM-L6-v2) student–reference cosine, mean absolute embedding difference, and mean Hadamard product.
- **Branch B — linguistic (23):** token/bigram/trigram overlap, precision/recall/Dice variants, content-word overlap on the negation-marked view, length and length-ratio statistics, TF–IDF cosine, negation cue counts and student–reference polarity mismatch, and named-entity overlap statistics.
- **Branch C — rubric coverage (5):** each reference-answer sentence is treated as a rubric concept; per-concept max SBERT similarity to the student answer yields mean/min/max similarity, concept count, and `coverage@ $\tau$` ($\tau=0.5$, fixed, untuned).

Reference-dependent features are NaN where no reference exists. The full feature dictionary (definition, view, NaN conditions per feature) ships with the code.

4.3 Fusion head

The head is LightGBM [40]: a classifier for categorical targets and a regressor otherwise; ordinal targets use regression followed by rounding and clipping to the training grade range. LightGBM handles NaN natively at split time, which is what makes a single 31-feature space usable across datasets with and without reference answers. ASAP-SAS is evaluated per prompt (one model per EssaySet, QWK macro-averaged), since each prompt has its own rubric and scale.

4.4 Hybrid arm: transformer-as-feature

The hybrid does *not* replace the head. A DeBERTa-v3-base [41, 42] cross-encoder is fine-tuned on (reference-or-question, student answer) pairs with a task-appropriate head, and its predictions are exported *out of fold*: for grouped-CV datasets, each fold’s rows are scored by a model trained on the other folds; for official-split datasets, test rows are scored by a model trained on the train split, while train rows receive inner grouped-CV out-of-fold scores so the fusion head never sees in-sample neural predictions. The exported signals (`neural_score`, `neural_pred`) are concatenated to the

feature matrix as branch D, preserving the NaN-native head and exact TreeSHAP attributions—including the ability to measure how much attribution mass the neural signal absorbs from the interpretable branches. [TBD: This arm is fully implemented; numbers await the GPU run (Section 7.9).]

4.5 Hyperparameter optimization and controls

Per dataset, Optuna [43] (TPE, 40 trials, fixed sampler seed) tunes seven LightGBM parameters (learning rate, leaves, estimators, child samples, subsample, column subsample, ℓ_1/ℓ_2) on the official dev split when one exists, otherwise on an inner grouped CV carved from training rows only—**the objective never touches a test split**. Every learned result is paired with two controls: a *trivial baseline* (majority/mean/median of train) and a *question-shortcut* model that predicts each test answer using only the training mean (or majority label) of its question, backing off to the global statistic for unseen questions. The question-shortcut arm is the leakage diagnosis made into a model: any score it achieves is obtainable without reading the student’s answer.

5 Leakage-Aware Evaluation Protocol

Grouped leave-questions-out CV. For datasets without official splits we use stratified group k -fold ($k=5$) with `question_id` as the group and binned scores as strata, so no question ever spans train and test. Folds are materialized once and reused by every experiment (training, HPO inner loops, ablations, XAI), and 5 seeds are reported as mean $\pm$ std.

Seen-question upper bound and generalization gap. Alongside the grouped estimate we compute the legacy stratified k -fold as a *transient seen-question upper bound*; the difference is the leakage the grouped protocol removes (RQ1). For official-split datasets the analogous gap is the drop from unseen-answer to unseen-question/domain test sets.

Cluster-bootstrap significance with family-wise control. Item-level bootstraps assume independent items, but answers to one question are correlated; resampling items overstates the effective sample size exactly on the datasets with few questions. We therefore resample *whole questions* with replacement (10,000 replicates) and report the observed $\Delta = \text{head} - \text{baseline}$, its percentile 95% CI, and the one-sided $p = P(\Delta \leq 0)$, Holm-corrected across the six datasets [35, 36]. For the per-prompt ASAP-SAS, the prompt is the cluster and answers are resampled within prompt. Significance is computed on the tuned head’s seed-42 predictions; multi-seed variability is reported separately in Table 3 (it is an order of magnitude smaller than the CIs).

Human ceiling. ASAP-SAS retains a second rater on train+dev; the inter-annotator QWK (per prompt 0.914–0.956, macro 0.942) contextualizes machine QWK as a corpus/rubric property. The other corpora do not retain per-rater reads, which we report as a limitation rather than approximating a ceiling.

6 Experimental Setup

All experiments are CPU-only (feature slice) with global seed 42; results report 5 seeds (42, 1, 2, 3, 4) as mean $\pm$ std. With default (untuned) LightGBM parameters the head is deterministic (subsample and column subsample = 1.0), so the 5-seed std of the untuned configuration is exactly

Dataset	Metric	Full head		Question-shortcut		η^2
		strat. (leaky)	grouped	strat. (leaky)	grouped	
Mohler	Pearson	0.570	0.430	0.365	-0.166	0.186
Powergrading	macro-F1	0.830	0.450	0.873	0.392	0.915
MIND-CA	QWK	0.121	0.059	0.164	0.000	0.141

Table 2: Question-leakage audit (default head; `reports/phase4_audit/`). “Strat.” = stratified k -fold with questions spanning folds (leaky); “grouped” = leave-questions-out. The question-shortcut model predicts from question identity alone. η^2 = between-question share of score variance. A lightly regularized head replicates the pattern (inflation +0.17, +0.35, +0.08 respectively).

zero; tuned configurations subsample and therefore exhibit honest non-zero seed variance. Python 3.11; LightGBM, scikit-learn [44], pandas [45], spaCy, sentence-transformers; 73 unit tests cover schema, split, feature, metric, and head contracts. Dataset downloads are checksummed; all reports quoted here are regenerated by single `make` targets.

7 Results

7.1 RQ1: The question-leakage audit

Table 2 quantifies the inflation on the three corpora where stratified k -fold is common practice. Two observations. First, the inflation is large and dataset-dependent: Powergrading loses 38 macro-F1 points when questions are held out. Second, the *question-shortcut control explains the inflation*: under the leaky protocol, a model that sees only `question_id` nearly matches (Mohler: 0.365 vs. 0.570) or even *exceeds* (Powergrading: 0.873 vs. 0.830) the full 31-feature model, while under the grouped protocol it collapses to (or below) the trivial baseline. The between-question variance share predicts vulnerability: Powergrading’s labels are almost entirely question-determined ($\eta^2=0.92$), MIND-CA’s much less so ($\eta^2=0.14$), and the inflation tracks this ordering. The official-split corpora corroborate the audit. On SemEval, the tuned head drops from 0.508 (unseen answers) to 0.453 (unseen questions) and 0.427 (unseen domains), while the question-shortcut arm scores 0.323 macro-F1 on unseen answers—nearly triple the majority baseline (0.118)—and collapses to the baseline on unseen questions, as it must. SAF is the extreme case: seen questions yield Pearson $r=0.904$, of which the question shortcut alone achieves 0.891; on unseen questions the head falls to 0.019 ± 0.033 , i.e., essentially nothing the model learned about SAF’s five test-UQ questions transfers. We therefore treat SAF as an explainability case study (Section 7.5) rather than an accuracy result.

7.2 Main results under the honest protocol

Table 3 reports the tuned head against both controls on each dataset’s headline split. Tuning adds modest, consistent gains over the default head (e.g., SemEval $0.411 \rightarrow 0.427$; Mohler $0.430 \rightarrow 0.490$; Powergrading $0.450 \rightarrow 0.486$). The ASAP-SAS QWK of 0.363 should be read against both the human ceiling (0.942) and the fact that only branch B is populated (no reference answers); the remaining gap is the direct measurement of how far interpretable answer-side features alone are from rubric-informed human judgment on this subset.

Dataset (headline split)	Metric	Tuned head	Default head	Trivial base	Q-shortcut
SemEval (test_ud)	macro-F1	0.427 ± 0.005	0.411	0.118	0.118
ASAP-SAS (test_ua)	QWK	0.363 ± 0.003	0.350	0.000	0.000
Mohler (grouped CV)	Pearson	0.490 ± 0.001	0.430	—	−0.166
SAF (test_uq)	Pearson	0.019 ± 0.033	0.026	—	—
Powergrading (grouped CV)	macro-F1	0.486 ± 0.009	0.450	0.392	0.392
MIND-CA (grouped CV)	QWK	0.077 ± 0.003	0.059	0.000	0.000

Table 3: Tuned head (Optuna, mean $\pm$ std over 5 seeds) vs. default head and controls on headline splits (**reports/phase2d/**). “—” = degenerate (constant) predictor, correlation undefined. On k-fold datasets the trivial baseline and question-shortcut coincide by construction on unseen questions.

Dataset	Effect	Δ	95% CI	p_{Holm}	significant
SemEval	head − base	0.315	[0.257, 0.361]	$< 10^{-3}$	yes
ASAP-SAS	head − base	0.360	[0.318, 0.398]	$< 10^{-3}$	yes
Mohler	head − q-shortcut	0.658	[0.463, 0.864]	$< 10^{-3}$	yes
MIND-CA	head − base	0.073	[0.017, 0.136]	0.011	yes
Powergrading	head − q-shortcut	0.079	[−0.069, 0.202]	0.291	no
SAF	head vs. zero	−0.018	[−0.072, 0.131]	0.582	no

Table 4: Cluster bootstrap over questions (10k replicates, one-sided $p=P(\Delta \leq 0)$, Holm-corrected across six tests; **reports/phase2d/significance.json**). ASAP-SAS resamples answers within prompt. SAF’s trivial baseline is a constant (correlation undefined), so the test is head-vs-zero. Point estimates use seed-42 predictions.

7.3 Significance under the cluster bootstrap

Table 4 and Figure 1 give the cluster-bootstrap analysis. The head’s gain is significant on SemEval, ASAP-SAS, Mohler, and MIND-CA, and *not* significant on Powergrading ($\Delta=0.079$, $p_{\text{Holm}}=0.291$; only 20 questions) or SAF ($p=0.582$; 5 test questions). An item-level bootstrap would have declared Powergrading significant; resampling questions is what prevents that error. This is the statistical face of the audit in Section 7.1: with few questions and high η^2 , both point estimates and their uncertainties are dominated by *which questions you have*, not how many answers.

7.4 RQ2: Branch ablations under the honest protocol

Table 5 removes or isolates each branch with one fixed, lightly regularized head (so differences are attributable to features, not tuning). Three findings. **(i) Linguistic features are the workhorse where a reference answer exists or length/style signal is informative:** removing branch B costs −0.109 macro-F1 on SemEval, −0.340 QWK on ASAP-SAS (where B is the only populated branch), −0.068 Pearson on Mohler, and all of MIND-CA’s skill. **(ii) The semantic and rubric branches are accuracy-light:** −A costs at most 0.020 anywhere, and −C is within noise (−0.003 to +0.008); on Powergrading, similarity-only variants (only-A 0.483, only-C 0.487) actually edge out the full model (0.473), suggesting the linguistic branch mildly overfits question style there. **(iii) Negation-scope cues deliver small but consistent gains** on five of six datasets (−neg costs 0.009–0.029), validating the negation-aware preprocessing; the one exception (MIND-CA, +0.006) has answers about mental states where our clause-bounded window is likely too blunt. Overall: interpretability does not have to be paid for with accuracy—the faithful rubric branch rides along at no cost—but the 2019-era bi-encoder semantics contribute little, which is precisely the gap the

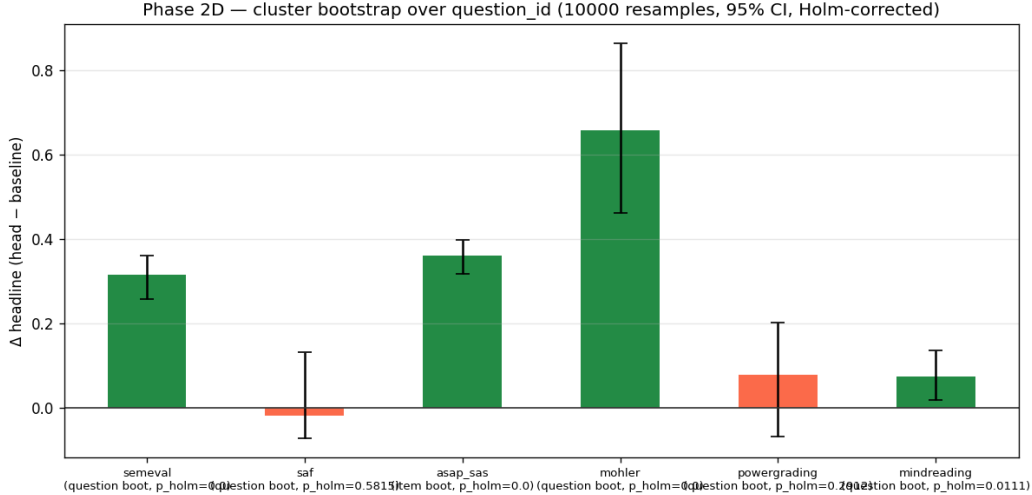


Figure 1: Per-dataset Δ with cluster-bootstrap 95% CIs; colored by whether the effect survives Holm correction.

Δ vs. full	SemEval	SAF	ASAP-SAS	Mohler	Powergr.	MIND-CA
full (abs.)	0.415	0.024	0.340	0.439	0.473	0.063
–A (semantic)	–0.014	–0.020	0.000	+0.000	–0.002	0.000
–B (linguistic)	–0.109	–0.027	–0.340	–0.068	+0.009	–0.063
–C (rubric)	–0.003	–0.007	0.000	+0.008	–0.001	0.000
–neg	–0.029	–0.021	–0.021	–0.021	–0.009	+0.006
only-A	–0.118	+0.008	–0.340	–0.076	+0.010	–0.063
only-B	–0.084	+0.022	0.000	+0.010	–0.056	0.000
only-C	–0.128	–0.192	–0.340	–0.105	+0.014	–0.063

Table 5: Branch ablations, fixed regularized head, mean over 5 seeds (reports/phase3/ablations.json); headline metric per dataset as in Table 3. ASAP-SAS and MIND-CA have no reference answer, so branches A/C are all-NaN and their removal is a no-op by construction.

hybrid arm targets. [TBD: revisit sentence after hybrid run]

7.5 RQ3: Explanations—faithful, and validated against humans

Global attribution sanity. Exact TreeSHAP global rankings agree with LightGBM gain rankings (Spearman ρ : SemEval 0.87, SAF 0.65, ASAP-SAS 0.998, Mohler 0.96, Powergrading 0.88, MIND-CA 0.999), so the explanation layer is not an artifact of one importance definition.

Faithfulness decomposition. For each feature we separate *predictive validity* (Spearman of feature vs. gold score), *model use* (Spearman of feature vs. its SHAP contribution), and *sign consistency* (fraction of items where the SHAP sign matches the feature’s deviation from its median). On Mohler, the rubric branch has the highest predictive validity of the three branches ($\rho=0.34$ vs. 0.20 linguistic, 0.14 semantic) while receiving little attribution mass (21% of global $|\text{SHAP}|$ vs. 72% linguistic)—i.e., the rubric signals are *valid but under-used* by the accuracy-optimal model. Combined with the $-C \approx 0$ ablation, this yields the paper’s central interpretability finding: **concept-**

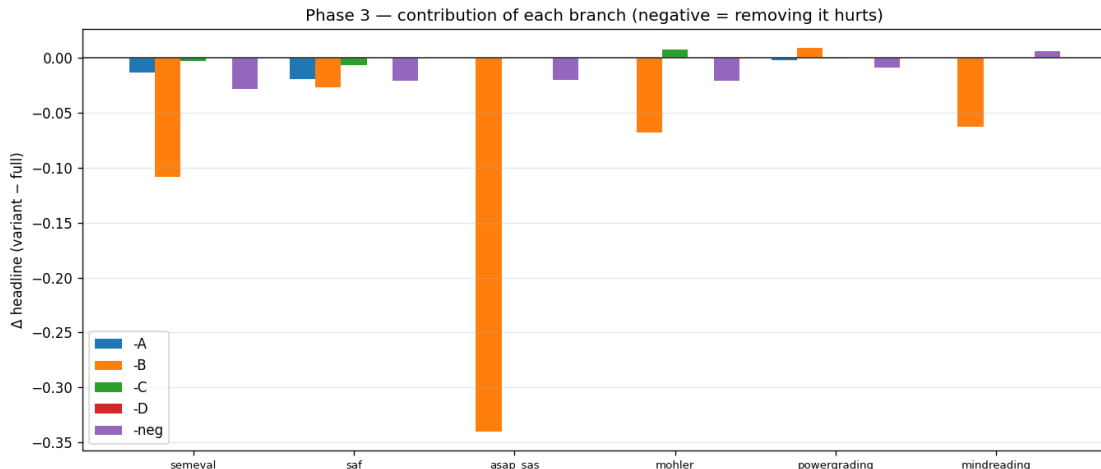


Figure 2: Branch ablation deltas vs. the full model (`reports/phase3/`).

Signal	Incorrect	Partial	Correct	Spearman ρ	AUC
rubric coverage@ τ	0.224	0.476	0.547	0.188	0.589
rubric mean max-sim	0.399	0.488	0.521	0.198	0.599
rubric min max-sim	0.306	0.347	0.391	0.139	0.575
semantic cosine	0.460	0.636	0.679	0.255	0.621

Table 6: SAF explanation validation: mean signal per human verdict class ($n=854$: 63 Incorrect / 304 Partially correct / 487 Correct), Spearman vs. the ordinal verdict, and AUC for Correct-vs-rest (`reports/phase2f/saf_validation.json`). All four signals are monotonic in the human verdict.

coverage explanations can be reported to teachers at essentially zero accuracy cost, but they do not (yet) buy accuracy—the head can reconstruct their information from cheaper lexical signals.

Concept-level attribution. Reconstructing per-concept coverage (each reference sentence a concept, covered iff max similarity $\geq \tau$) yields teacher-style explanations of the form “covered c_1, c_3 ; missed c_2 ”, e.g., for a SemEval electricity item, the student’s “positive battery terminal is separated by a gap from terminal 1” matches the single reference concept at similarity 0.79 and is explained as fully covered.

Validation against human gold feedback (SAF). SAF ships a human verification verdict per answer (Correct / Partially correct / Incorrect), which the unified schema discards and we recover ($n=854$ across both test splits). All four interpretable signals rise strictly monotonically across the verdict classes (Table 6); semantic cosine is the strongest ($\rho=0.255$, $\text{AUC}_{\text{Correct-vs-rest}}=0.621$). The effect is moderate, not decisive—but it is measured against *pre-existing* human feedback rather than annotations collected to fit the method, and it turns “the explanations look plausible” into a quantitative claim.

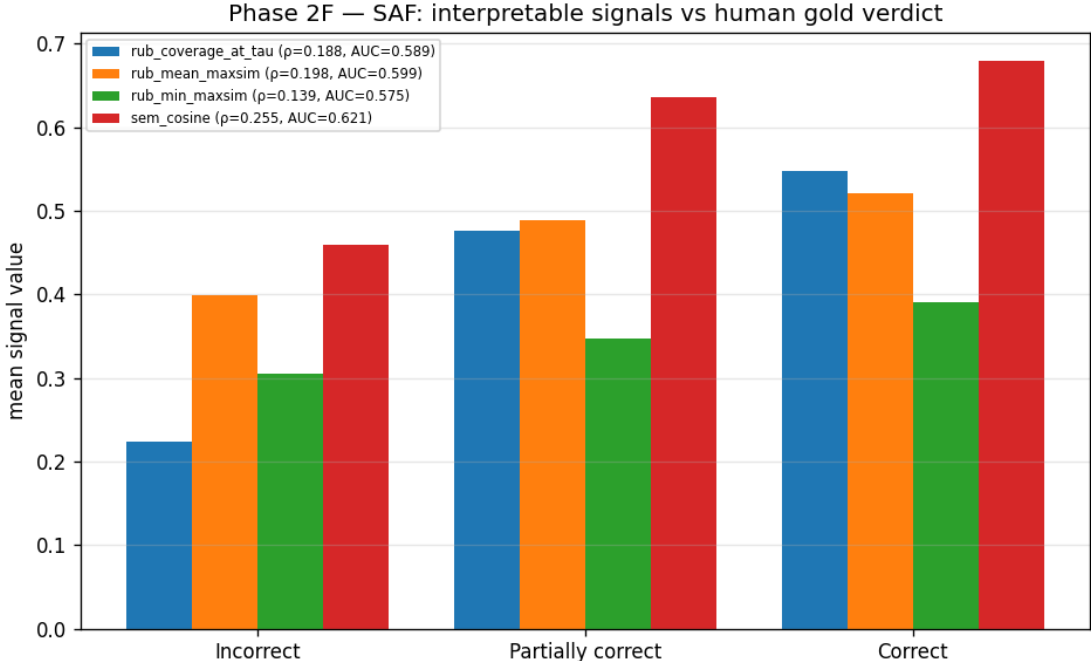


Figure 3: SAF: interpretable signals vs. human feedback class (`reports/phase2f/`).

7.6 RQ4a: Calibration and selective prediction

On SemEval (unseen domains), the raw head is overconfident (ECE 0.122); temperature scaling ($T=1.35$, fit on a held-out slice of training data, never on test) halves it to 0.059 (Figure 4). Powergrading under unseen questions is severely miscalibrated (ECE 0.613), and even the optimal temperature ($T=5.95$) only reduces it to 0.334—quantifying how little the confidence of a question-memorizing classifier means on new questions. Risk-coverage analysis supports deferral-based deployment [32]: routing the least-confident 20% of SemEval answers to a human raises accuracy on the remainder from 0.467 to 0.506, and to 0.575 at 50% coverage.

7.7 RQ4b: Perturbation robustness

Three text perturbations probe known failure modes [29, 31]: appending irrelevant filler (*length pad*), randomly dropping 30% of tokens (*paraphrase proxy*), and marking all content tokens as negated (*negation flip*). Only lexical/negation features are recomputed (semantic and rubric features held fixed), making the reported sensitivities a conservative bound. As intended, the grader reacts most strongly to negation flips on SemEval (49% of predictions change; mean absolute label shift 0.80)—a grader that ignores negation would be worse than a brittle one—while the paraphrase proxy changes 18% of SemEval predictions and systematically lowers Mohler scores (mean signed shift -0.24), exposing the expected lexical-overlap reliance of branch B. Length padding shifts predictions on 32% of SemEval items, confirming that length features remain a partially open attack surface even under grouped evaluation.

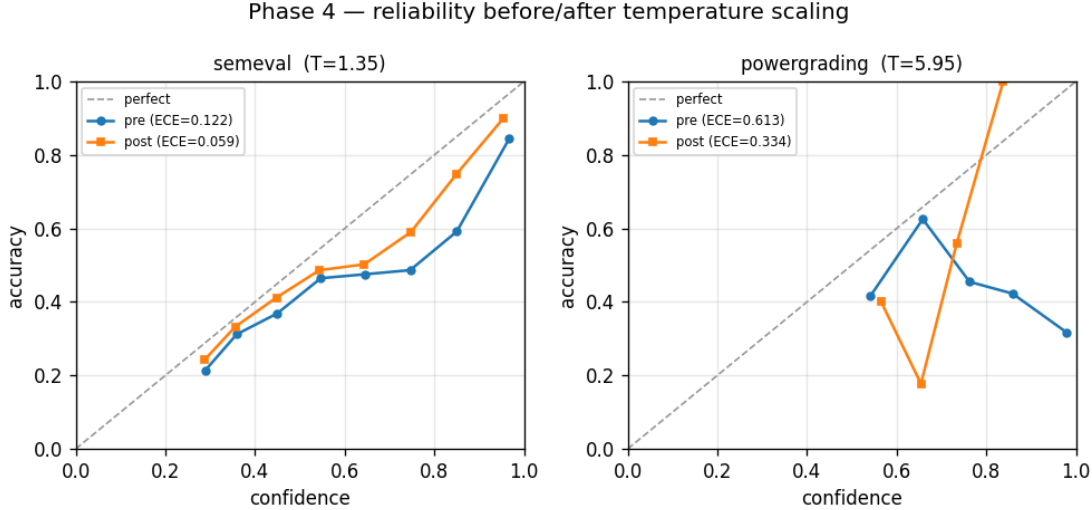


Figure 4: Reliability before/after temperature scaling (`reports/phase4_robust/`).

7.8 RQ4c: Cross-dataset transfer (LODO)

Mapping every dataset to a binary correct/incorrect target and training on five datasets’ shared 31 features while testing on the sixth yields weak and asymmetric transfer (macro-F1 0.24–0.63): useful signal flows into SemEval (0.61 macro-F1, AUC 0.68) and Powergrading (0.63, 0.71), SAF is chance-level (AUC 0.50), and ASAP-SAS *inverts* (AUC 0.27)—features that indicate correctness elsewhere anti-correlate with it there, plausibly because ASAP’s binarized rubric scores reward length-and-detail patterns that differ from the overlap-with-reference regime of the other corpora. We report this honest negative result as a caution against “train once, grade anywhere” claims for feature-based graders; it also mirrors recent findings that cross-domain ASAG transfer is fragile even for LLMs [22].

7.9 RQ5: Hybrid arm (pending GPU run)

The three-way comparison—neural-only cross-encoder, feature-only head, hybrid (features + out-of-fold neural signals)—under identical grouped folds is fully implemented; the DeBERTa fine-tuning requires a GPU and has not yet been executed. Table 7 will report it. **[TBD: Run notebooks/02_neural_colab.ipynb, then make neuralfeatures-dependent re-runs; fill the table; update the abstract, Sections 2.4, 7.4, 8, and the conclusion.]**

7.10 Context against published results

No number in this paper is split-, metric-, and corpus-matched to a published result, so we make no head-to-head claims: our Mohler is the 21-question ASAG2024 subset (published Pearson ~ 0.52 refers to the 80-question 2011 corpus [3]), and our ASAP-SAS covers 4 of 10 prompts (published neural QWK of 0.72–0.73 refers to all ten [14]). Our SemEval headline is the hardest split (unseen domains); published 5-way macro-F1 figures are typically reported on unseen answers. We provide our unseen-answer numbers (0.508 macro-F1) to enable future split-matched comparison, and release per-item predictions.

Dataset (headline)	Feature-only	Neural-only	Hybrid	Δ hybrid—feat
SemEval (test_ud, macro-F1)	0.427 ± 0.005	[TBD:]	[TBD:]	[TBD:]
SAF (test_uq, Pearson)	0.019 ± 0.033	[TBD:]	[TBD:]	[TBD:]
ASAP-SAS (test_ua, QWK)	0.363 ± 0.003	[TBD:]	[TBD:]	[TBD:]
Mohler (CV, Pearson)	0.490 ± 0.001	[TBD:]	[TBD:]	[TBD:]
Powergrading (CV, macro-F1)	0.486 ± 0.009	[TBD:]	[TBD:]	[TBD:]
MIND-CA (CV, QWK)	0.077 ± 0.003	[TBD:]	[TBD:]	[TBD:]

Table 7: [TBD: Hybrid three-way comparison under identical grouped folds (reports/phase_hybrid/three_way.json after the GPU run). Also to report: share of global |SHAP| absorbed by neural_* (explanation dilution), and the −D ablation row.]

8 Discussion

What the audit means for the field. The single largest determinant of a reported ASAG number on no-official-split corpora is not the model but *whether questions span folds*. Under the leaky protocol, question identity alone is a top-tier “model” (Table 2). This has two practical consequences: published stratified-CV results on Mohler/Powergrading-style corpora should be read as seen-question upper bounds, and new-dataset authors should ship grouped or unseen-question splits (as SemEval and SAF already do). The cluster bootstrap is the inferential counterpart: on 20-question corpora, answer-level CIs are illusions of precision.

What carries accuracy, honestly. After leakage is removed, hand-crafted linguistic features still carry the model; 2019-era bi-encoder semantics add little; and the rubric branch adds explanation without accuracy. We consider the last point a feature, not a failure: the coverage signals are valid (they track grades and human verdicts), so the accuracy-optimal tree simply prefers cheaper correlates. Whether a *stronger* semantic signal changes this hierarchy is exactly RQ5. [TBD: one paragraph on the hybrid outcome]

Deployment posture. The full system is CPU-only, trains in minutes, is NaN-robust across schema variation, exposes exact attributions, and its confidence can be temperature- corrected and thresholded for deferral. In an era of LLM graders with known leniency and reliability concerns [27, 28], an auditable local grader with quantified unseen-question behavior is a practical alternative for institutions that must justify grades.

9 Error Analysis

On SemEval unseen domains (exact accuracy 46.5%), the dominant confusion is `partially_correct_incomplete` predicted as `correct` (524 of 986 partials; class F1 0.267)—overlap features cannot reliably distinguish “mentions the right concept” from “mentions it completely”, which is precisely the granularity the rubric branch was designed for and where a cross-encoder should help most. The second confusion, `irrelevant`→`correct` (490 of 1,222; F1 0.350), reflects topically-related but non-responsive answers defeating bag-of-similarity features. `contradictory` recall is 0.213: negation cues alone under-detect contradiction, consistent with the small-but-real −neg ablation. The class-frequency skew matters too: `non_domain` has 20 test items (F1 0.625 on trivial support). On Mohler, the perturbation analysis shows score under-prediction under token drop (−0.24 signed), confirming

recall-oriented overlap reliance. These patterns jointly predict where the hybrid should pay: partial-vs-correct boundaries and contradiction detection. [TBD: verify prediction post-run]

10 Threats to Validity

Internal. HPO for k-fold datasets uses inner CV on training rows; a full nested-CV protocol was not run (documented mild optimism; the tuned-vs- default gaps in Table 3 bound its size). Significance point estimates use seed-42 predictions; 5-seed stds are reported and are an order of magnitude smaller than bootstrap CIs. Perturbation results recompute only lexical/negation features (conservative) and are single-run point estimates.

Construct. Reference sentences are a rubric *proxy*, not authored rubrics; coverage@ τ uses one fixed $\tau=0.5$ (untuned by design). The unified binary target in LODO imposes a midpoint threshold on ordinal/continuous scales.

External. English only; short answers only; Mohler and ASAP-SAS are subsets of their original corpora; SAF test-UQ has five questions, so its unseen-question estimate is high-variance.

Statistical. With 11–31 questions per corpus, cluster-bootstrap CIs are wide by construction; we prefer honest width to item-level optimism.

11 Limitations

(1) The semantic branch uses a lightweight 2019-era bi-encoder; its weak contribution is a statement about that encoder, not about semantics—the pending DeBERTa arm addresses this directly. (2) The rubric proxy degenerates for single-sentence references. (3) No human study evaluates whether the concept-coverage explanations are *useful* to teachers; we validate alignment with human verdicts, not pedagogical utility. (4) No LLM baseline is included; the protocol applies to LLM graders unchanged and such a row is planned. (5) Runtime/memory profiles, learning curves, and a negation-window sensitivity analysis are not yet reported. (6) Datasets are modest in size and the number of questions per corpus (11–252) bounds the precision of any unseen-question estimate.

12 Future Work

Immediate: execute the DeBERTa hybrid (RQ5) and the $-D$ /only- D ablations; add a zero-shot LLM row under the identical protocol; report runtime and learning curves. Near-term: rank-consistent ordinal heads (CORAL/CORN [46, 47]), which to our knowledge remain unapplied in ASAG; widening ASAP-SAS to all ten prompts and Mohler to the full 2011 corpus for split-matched comparison; a teacher-facing user study of concept-coverage explanations. Longer-term: multilingual extension (SAF is bilingual) and integrating the audit into dataset releases as a standard reporting artifact.

13 Conclusion

We presented a leakage-aware evaluation of interpretable ASAG across six heterogeneous datasets. Question-level leakage inflates the still-common stratified-CV protocol by up to 38 macro-F1 points,

and a question-identity-only control model explains the inflation; grouped leave-questions-out CV with cluster-bootstrap significance gives the honest picture, in which the fusion head beats its controls on four of six datasets. Hand-crafted linguistic features remain the accuracy workhorse; rubric concept-coverage explanations are faithful, align monotonically with human gold feedback on SAF, and cost no accuracy; calibration and deferral analyses show what it takes to deploy such a grader responsibly. The framework—grouped folds, shortcut controls, cluster bootstrap, explanation validation—applies unchanged to transformer and LLM graders, and the pending DeBERTa-as-feature arm will test whether stronger semantics change the accuracy hierarchy without sacrificing exact attributions. **[TBD: final sentence after hybrid run]**

Reproducibility and Data Availability

All code, unified splits (including materialized fold assignments), feature dictionaries, and every report JSON quoted in this paper are released at **[TBD: repository URL]**. Raw datasets are not redistributed; download scripts with checksums and license notes are provided. Global seed 42; 73 unit tests cover the schema, split, feature, metric, and head contracts.

References

- [1] A. T, 2020.
- [2] A. T, 2020.
- [3] A. T, 2020.
- [4] A. T, 2020.
- [5] A. T, 2020.
- [6] A. T, 2020.
- [7] A. T, 2020.
- [8] A. T, 2020.
- [9] A. T, 2020.
- [10] A. T, 2020.
- [11] A. T, 2020.
- [12] A. T, 2020.
- [13] A. T, 2020.
- [14] A. T, 2020.
- [15] A. T, 2020.
- [16] A. T, 2020.
- [17] A. T, 2020.

- [18] A. T, 2020.
- [19] A. T, 2020.
- [20] A. T, 2020.
- [21] A. T, 2020.
- [22] A. T, 2020.
- [23] A. T, 2020.
- [24] A. T, 2020.
- [25] A. T, 2020.
- [26] A. T, 2020.
- [27] A. T, 2020.
- [28] A. T, 2020.
- [29] A. T, 2020.
- [30] A. T, 2020.
- [31] A. T, 2020.
- [32] A. T, 2020.
- [33] A. T, 2020.
- [34] A. T, 2020.
- [35] A. T, 2020.
- [36] A. T, 2020.
- [37] A. T, 2020.
- [38] A. T, 2020.
- [39] A. T, 2020.
- [40] A. T, 2020.
- [41] A. T, 2020.
- [42] A. T, 2020.
- [43] A. T, 2020.
- [44] A. T, 2020.
- [45] A. T, 2020.
- [46] A. T, 2020.
- [47] A. T, 2020.

A Tuned hyperparameters

Optuna TPE, 40 trials per dataset, seed fixed; search space: learning rate $[0.005, 0.3]$ (log), leaves $[15, 255]$, estimators $[100, 800]$, child samples $[5, 80]$, subsample and column subsample $[0.6, 1.0]$, ℓ_1/ℓ_2 $[10^{-3}, 10]$ (log).

Dataset	lr	leaves	estim.	child	subsamp.	colsamp.	ℓ_1	ℓ_2
SemEval	0.088	114	300	80	0.906	0.691	9.94	0.23
SAF	0.096	23	350	17	0.927	0.807	0.07	0.72
ASAP-SAS	0.023	35	750	41	0.832	0.904	0.64	1.22
Mohler	0.015	115	350	74	0.636	0.893	1.66	0.02
Powergrading	0.229	231	650	48	0.790	0.749	5.18	5.86
MIND-CA	0.146	49	700	52	0.785	0.752	0.80	0.33

Table 8: Per-dataset tuned LightGBM parameters (`reports/phase2d/results.json`).

B SemEval per-class performance (unseen domains)

Per-class F1 (support): correct 0.633 (1,917); non_domain 0.625 (20); irrelevant 0.350 (1,222); partially_correct_incomplete 0.267 (986); contradictory 0.233 (417). Top confusions: partial→correct (524), irrelevant→correct (490), irrelevant→partial (279). Full confusion matrices for all datasets: `reports/phase3/error_analysis.json`.